

Building Technology

Engineering Mathematics II for Building Technology

Professional Learner Notes

Advanced algebra, calculus, vectors, matrices and construction data interpretation.

Section	Purpose
Advanced algebra	Equations, functions and construction formulas.
Calculus	Differentiation and integration in quantity and cost models.
Vectors	Components of loads and directions.
Matrices	Simultaneous equations and quantity planning.
Construction data	Statistics and quality interpretation.

How to Study This Topic

The learning method follows a programmed engineering mathematics approach: understand the concept, study the formula, follow a worked method, then practise with a technical problem. The aim is not memorisation only; the learner must interpret the answer in the department context.

- Write the formula before substitution.
- Use correct units.
- Check whether the result is reasonable.
- State a technical interpretation.
- Repeat practice until the method is fluent.

Formula Summary

Derivative

$$\frac{dy}{dx} = f'(x)$$

Meaning: identify each variable, check the units, substitute carefully, and state the final answer in the correct engineering context.

Definite integral

$$\int_a^b f(x) dx$$

Meaning: identify each variable, check the units, substitute carefully, and state the final answer in the correct engineering context.

Vector resultant

$$\vec{R} = \vec{A} + \vec{B}$$

Meaning: identify each variable, check the units, substitute carefully, and state the final answer in the correct engineering context.

Matrix system

$$AX = B$$

Meaning: identify each variable, check the units, substitute carefully, and state the final answer in the correct engineering context.

Standard deviation

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n-1}}$$

Meaning: identify each variable, check the units, substitute carefully, and state the final answer in the correct engineering context.

1. Advanced algebra

Equations, functions and construction formulas.

In building technology, this section is important because trainees use the calculation to make practical decisions in drawings, site measurements, setting out, material estimation, survey observations, quality checks, or revision work.

Key learning points

- Identify the physical quantity being calculated.
- Separate given values from required values.
- Select a formula that matches the problem.
- Use consistent units before calculation.
- Interpret the result as a technician, not only as a number.

Main formula for this section

$$\frac{dy}{dx} = f'(x)$$

Worked method

Example: Cost optimization. Differentiate a quadratic cost function and find its stationary point.

Step 1: Write the known quantities. Step 2: Write the formula. Step 3: Substitute the values.

Step 4: simplify carefully. Step 5: attach units. Step 6: write a one-sentence technical interpretation.

Engineering application

This method supports cost optimization. The result can be used to check whether a site measurement, drawing interpretation, material quantity, or survey computation is technically reasonable.

Common learner errors

- Using the wrong formula because the diagram/context was not analysed.
- Mixing units such as mm, cm and m.
- Rounding too early.
- Losing signs or angle direction.
- Submitting an answer without interpretation.

Practice set

1. Create and solve one technical calculation involving matrix planning. Show formula, substitution, units and interpretation.
2. Create and solve one technical calculation involving material-test data. Show formula, substitution, units and interpretation.
3. Create and solve one technical calculation involving estimation checks. Show formula, substitution, units and interpretation.
4. Create and solve one technical calculation involving cost optimization. Show formula, substitution, units and interpretation.
5. Create and solve one technical calculation involving load components. Show formula, substitution, units and interpretation.

6. Create and solve one technical calculation involving matrix planning. Show formula, substitution, units and interpretation.
7. Create and solve one technical calculation involving material-test data. Show formula, substitution, units and interpretation.
8. Create and solve one technical calculation involving estimation checks. Show formula, substitution, units and interpretation.

2. Calculus

Differentiation and integration in quantity and cost models.

In building technology, this section is important because trainees use the calculation to make practical decisions in drawings, site measurements, setting out, material estimation, survey observations, quality checks, or revision work.

Key learning points

- Identify the physical quantity being calculated.
- Separate given values from required values.
- Select a formula that matches the problem.
- Use consistent units before calculation.
- Interpret the result as a technician, not only as a number.

Main formula for this section

$$\int_a^b f(x) dx$$

Worked method

Example: Load components. Resolve a load into horizontal and vertical components.

Step 1: Write the known quantities. Step 2: Write the formula. Step 3: Substitute the values.

Step 4: simplify carefully. Step 5: attach units. Step 6: write a one-sentence technical interpretation.

Engineering application

This method supports load components. The result can be used to check whether a site measurement, drawing interpretation, material quantity, or survey computation is technically reasonable.

Common learner errors

- Using the wrong formula because the diagram/context was not analysed.
- Mixing units such as mm, cm and m.
- Rounding too early.
- Losing signs or angle direction.
- Submitting an answer without interpretation.

Practice set

1. Create and solve one technical calculation involving material-test data. Show formula, substitution, units and interpretation.
2. Create and solve one technical calculation involving estimation checks. Show formula, substitution, units and interpretation.
3. Create and solve one technical calculation involving cost optimization. Show formula, substitution, units and interpretation.
4. Create and solve one technical calculation involving load components. Show formula, substitution, units and interpretation.
5. Create and solve one technical calculation involving matrix planning. Show formula, substitution, units and interpretation.

6. Create and solve one technical calculation involving material-test data. Show formula, substitution, units and interpretation.
7. Create and solve one technical calculation involving estimation checks. Show formula, substitution, units and interpretation.
8. Create and solve one technical calculation involving cost optimization. Show formula, substitution, units and interpretation.

3. Vectors

Components of loads and directions.

In building technology, this section is important because trainees use the calculation to make practical decisions in drawings, site measurements, setting out, material estimation, survey observations, quality checks, or revision work.

Key learning points

- Identify the physical quantity being calculated.
- Separate given values from required values.
- Select a formula that matches the problem.
- Use consistent units before calculation.
- Interpret the result as a technician, not only as a number.

Main formula for this section

$$\vec{R} = \vec{A} + \vec{B}$$

Worked method

Example: Quantity equations. Use simultaneous equations to solve material quantities.

Step 1: Write the known quantities. Step 2: Write the formula. Step 3: Substitute the values.

Step 4: simplify carefully. Step 5: attach units. Step 6: write a one-sentence technical interpretation.

Engineering application

This method supports matrix planning. The result can be used to check whether a site measurement, drawing interpretation, material quantity, or survey computation is technically reasonable.

Common learner errors

- Using the wrong formula because the diagram/context was not analysed.
- Mixing units such as mm, cm and m.
- Rounding too early.
- Losing signs or angle direction.
- Submitting an answer without interpretation.

Practice set

1. Create and solve one technical calculation involving estimation checks. Show formula, substitution, units and interpretation.
2. Create and solve one technical calculation involving cost optimization. Show formula, substitution, units and interpretation.
3. Create and solve one technical calculation involving load components. Show formula, substitution, units and interpretation.
4. Create and solve one technical calculation involving matrix planning. Show formula, substitution, units and interpretation.
5. Create and solve one technical calculation involving material-test data. Show formula, substitution, units and interpretation.

6. Create and solve one technical calculation involving estimation checks. Show formula, substitution, units and interpretation.
7. Create and solve one technical calculation involving cost optimization. Show formula, substitution, units and interpretation.
8. Create and solve one technical calculation involving load components. Show formula, substitution, units and interpretation.

4. Matrices

Simultaneous equations and quantity planning.

In building technology, this section is important because trainees use the calculation to make practical decisions in drawings, site measurements, setting out, material estimation, survey observations, quality checks, or revision work.

Key learning points

- Identify the physical quantity being calculated.
- Separate given values from required values.
- Select a formula that matches the problem.
- Use consistent units before calculation.
- Interpret the result as a technician, not only as a number.

Main formula for this section

$$AX=B$$

Worked method

Example: Cost optimization. Differentiate a quadratic cost function and find its stationary point.

Step 1: Write the known quantities. Step 2: Write the formula. Step 3: Substitute the values.

Step 4: simplify carefully. Step 5: attach units. Step 6: write a one-sentence technical interpretation.

Engineering application

This method supports material-test data. The result can be used to check whether a site measurement, drawing interpretation, material quantity, or survey computation is technically reasonable.

Common learner errors

- Using the wrong formula because the diagram/context was not analysed.
- Mixing units such as mm, cm and m.
- Rounding too early.
- Losing signs or angle direction.
- Submitting an answer without interpretation.

Practice set

1. Create and solve one technical calculation involving cost optimization. Show formula, substitution, units and interpretation.
2. Create and solve one technical calculation involving load components. Show formula, substitution, units and interpretation.
3. Create and solve one technical calculation involving matrix planning. Show formula, substitution, units and interpretation.
4. Create and solve one technical calculation involving material-test data. Show formula, substitution, units and interpretation.
5. Create and solve one technical calculation involving estimation checks. Show formula, substitution, units and interpretation.

6. Create and solve one technical calculation involving cost optimization. Show formula, substitution, units and interpretation.
7. Create and solve one technical calculation involving load components. Show formula, substitution, units and interpretation.
8. Create and solve one technical calculation involving matrix planning. Show formula, substitution, units and interpretation.

5. Construction data

Statistics and quality interpretation.

In building technology, this section is important because trainees use the calculation to make practical decisions in drawings, site measurements, setting out, material estimation, survey observations, quality checks, or revision work.

Key learning points

- Identify the physical quantity being calculated.
- Separate given values from required values.
- Select a formula that matches the problem.
- Use consistent units before calculation.
- Interpret the result as a technician, not only as a number.

Main formula for this section

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n-1}}$$

Worked method

Example: Load components. Resolve a load into horizontal and vertical components.

Step 1: Write the known quantities. Step 2: Write the formula. Step 3: Substitute the values.

Step 4: simplify carefully. Step 5: attach units. Step 6: write a one-sentence technical interpretation.

Engineering application

This method supports estimation checks. The result can be used to check whether a site measurement, drawing interpretation, material quantity, or survey computation is technically reasonable.

Common learner errors

- Using the wrong formula because the diagram/context was not analysed.
- Mixing units such as mm, cm and m.
- Rounding too early.
- Losing signs or angle direction.
- Submitting an answer without interpretation.

Practice set

1. Create and solve one technical calculation involving load components. Show formula, substitution, units and interpretation.
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5. Create and solve one technical calculation involving cost optimization. Show formula, substitution, units and interpretation.

6. Create and solve one technical calculation involving load components. Show formula, substitution, units and interpretation.
7. Create and solve one technical calculation involving matrix planning. Show formula, substitution, units and interpretation.
8. Create and solve one technical calculation involving material-test data. Show formula, substitution, units and interpretation.

Mixed Revision Questions

1. A building technology problem involves load components. Choose a suitable mathematical method, solve the problem, and explain the result.
2. A building technology problem involves matrix planning. Choose a suitable mathematical method, solve the problem, and explain the result.
3. A building technology problem involves material-test data. Choose a suitable mathematical method, solve the problem, and explain the result.
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24. A building technology problem involves estimation checks. Choose a suitable mathematical method, solve the problem, and explain the result.
25. A building technology problem involves cost optimization. Choose a suitable mathematical method, solve the problem, and explain the result.
26. A building technology problem involves load components. Choose a suitable mathematical method, solve the problem, and explain the result.
27. A building technology problem involves matrix planning. Choose a suitable mathematical method, solve the problem, and explain the result.
28. A building technology problem involves material-test data. Choose a suitable mathematical method, solve the problem, and explain the result.
29. A building technology problem involves estimation checks. Choose a suitable mathematical method, solve the problem, and explain the result.
30. A building technology problem involves cost optimization. Choose a suitable mathematical method, solve the problem, and explain the result.

References and Learning Sources

These learner materials are original notes prepared for Mr Mwirigi Mathematics Training Hub. The topic sequencing and level are informed by standard engineering mathematics references and the uploaded curriculum/OS materials. No textbook pages or solution-manual pages have been reproduced.

1. John Bird, Engineering Mathematics, Newnes / Elsevier.
2. K. A. Stroud and Dexter J. Booth, Engineering Mathematics, Palgrave Macmillan.
3. Uploaded KNEC solved mathematics materials for examination-style orientation.
4. Civil Engineering, Building Technology and Land Survey curriculum and occupational-standard materials used for local alignment.